Red Hat
OpenShift

OpenShift on OpenShift using Hosted Control Planes and OpenShift Virtualization

Red Hat

Why OpenShift virtualization + HCP?

Customers want alternatives to VMware

Red Hat considers new virtualization deployments to be a critical business success factor

You want to be a trusted advisor to both your customer and a hero to Red Hat

Why OpenShift virtualization + HCP?

Customers want to see OpenShift virtualization in action, in a cost-effective way so that they can be a hero in their organizations

Virtualization is deployed on (expensive) bare-metal servers, which means that the OpenShift control plane consumes 3 very expensive bare-metal servers

You can guide them by reposing architectures that meet your customer’s need for efficiency and cost control

Business Value

OpenShift Control Plane Overhead

Openshift clusters require 3 master nodes

Control plane overhead is a huge concern for many customers considering OpenShift virtualization (on bare metal)

Control plane overhead = 3 / <Total Number of Nodes in the Cluster>

Example: for a 30-node cluster, control plane overhead is 10%

This level of overhead is visible to customers. They do not like it, and it presents a blocker to closing opportunities.

Running the Numbers

Consider This Scenario:

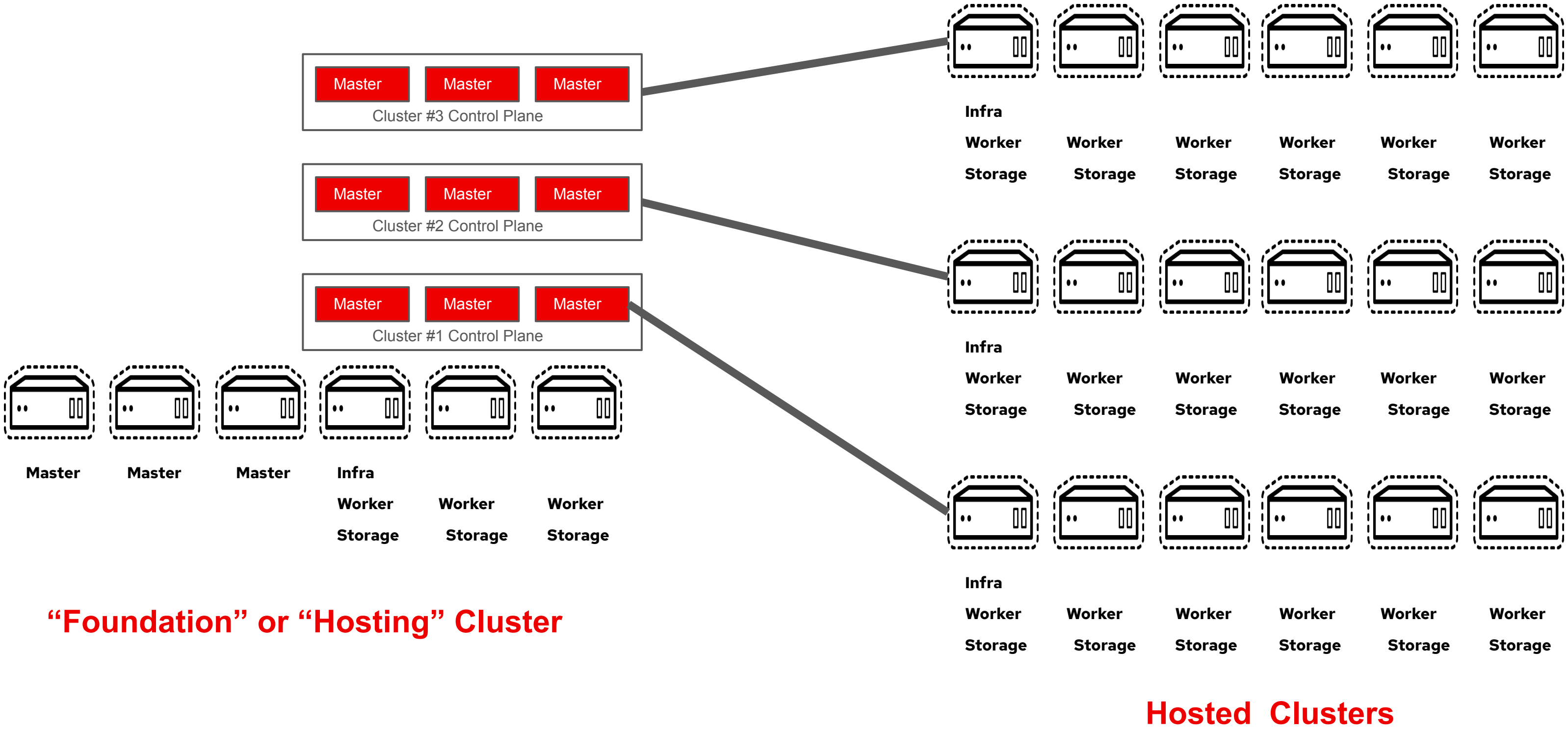
Hosting Cluster Size: 6 nodes (3 control plane, 3 workers)

Hosting Cluster Capacity: 21 hosted clusters (3 workers x 7 hosted clusters per worker)

Hosted Cluster Control Plane Nodes saved: 63
Hosting Cluster Nodes Used: 6
Reduction in Control Plane HW Cost: >90%

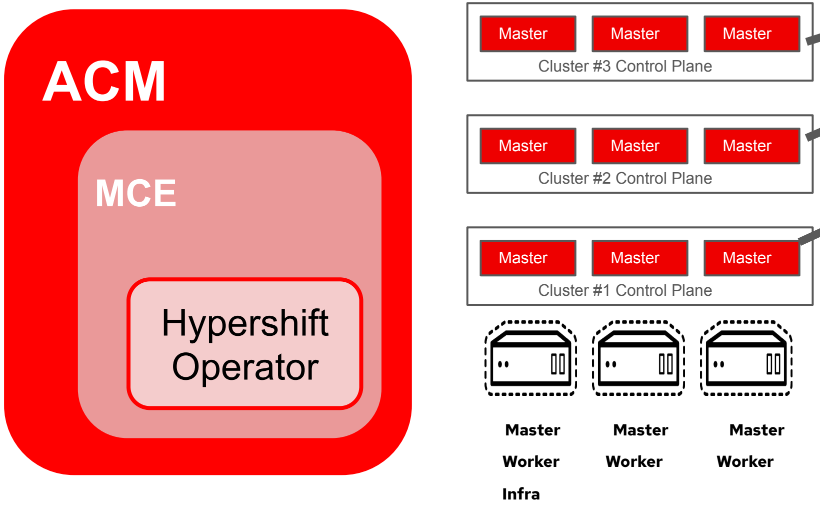
OpenShift cluster architecture

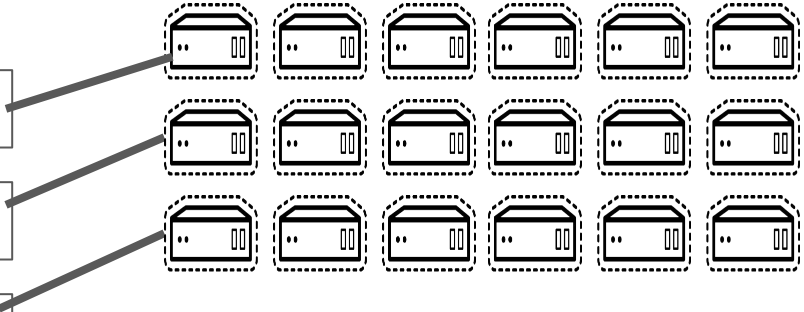
Hosted Control planes



OpenShift cluster architecture

Automating Multi-Cluster Deployment & Management





ACM (Advanced Cluster Manager)

ACM / HCP Integration

- ACM uses HCP to rapidly deploy OpenShift clusters
- ACM also consolidates telemetry, security, monitoring, & application deployment for multiple clusters
- ACM is not required; MCE (Multi-Cluster Engine) component can run standalone

Red Hat
OpenShift

Thank You!